

TASK 4: DASHBOARD DESIGN

Sales & Financial Performance Dashboard

Objective: Design an interactive dashboard that helps business stakeholders understand sales, profit, growth, and trends.

Student Name	Arshiya Chaudhary
Course	B.Sc. Data Science
Task	Task 4 – Dashboard Design
Tools	Tableau

1. Objective

The objective of this task is to design an interactive business dashboard using a Sales_Financial dataset. The dashboard should convert raw sales data into clear KPIs, trends, comparisons, and actionable insights. Users should be able to filter the dashboard by time, region, category, segment, and other available dimensions.

2. Dataset

The recommended dataset is a Sales_Financial dataset containing sales transactions and financial measures. If the Kaggle version has different column names, map the equivalent columns to the fields below.

Field	Type	Dashboard Use
Order Date	Date	Time-series analysis
Sales	Numeric	Total Sales KPI
Profit	Numeric	Total Profit KPI
Quantity	Numeric	Units sold
Category	Text	Category analysis
Sub-Category	Text	Detailed product analysis
Region	Text	Regional comparison
Customer / Segment	Text	Customer analysis

3. Dashboard KPIs

The dashboard should place the most important business KPIs at the top using card visuals.

KPI	Calculation	Business Meaning
Total Sales	SUM(Sales)	Shows overall revenue/sales generated.
Total Profit	SUM(Profit)	Shows overall profitability.
Total Quantity	SUM(Quantity)	Shows units sold.
Profit Margin	Profit ÷ Sales	Measures profitability relative to sales.
Sales Growth %	Current period vs previous period	Shows whether sales are increasing or decreasing.

4. Recommended Dashboard Layout

A clean one-page dashboard can be arranged as follows:

- Top header: Dashboard title and reporting period.

- KPI cards: Total Sales, Total Profit, Total Quantity, Profit Margin, Sales Growth %.
- Left or top slicers: Date, Region, Category, Segment.
- Main trend chart: Sales and Profit by Month.
- Category chart: Sales and Profit by Category/Sub-Category.
- Regional chart: Sales and Profit by Region.
- Top products/customers table: Ranked by Sales or Profit.
- Optional navigation buttons: Overview, Product Analysis, Regional Analysis.

5. Power BI Implementation – Step by Step

1. Open Power BI Desktop.
2. Select Home → Get Data → Text/CSV or Excel and import the Sales_Financial dataset.
3. Open Power Query and check data types, missing values, duplicate records, and date columns.
4. Rename fields if necessary so the calculations match the DAX measures below.
5. Create a Date table for reliable time-series and year/month analysis.
6. Create the KPI measures using DAX.
7. Add Card visuals for Sales, Profit, Quantity, Profit Margin, and Sales Growth.
8. Add slicers for Date, Region, Category, and Segment.
9. Add a line chart for monthly Sales and Profit.
10. Add bar/column charts for Category and Region comparisons.
11. Add a table or matrix for top products/customers.
12. Apply one consistent theme and readable formatting.
13. Add page navigation buttons if multiple dashboard pages are created.
14. Test every slicer and visual interaction before publishing/exporting.

6. Tableau Implementation

The same dashboard can be created in Tableau if Tableau is selected instead of Power BI.

- Connect Tableau to the Sales_Financial CSV/Excel file.
- Check data types and convert Order Date to a Date field.
- Create calculated fields for Profit Margin and Growth.
- Create worksheets for KPIs, monthly trends, category performance, regional performance, and top products.
- Add filters for Date, Region, Category, and Segment.
- Use Dashboard → New Dashboard to combine worksheets.
- Enable filter actions so selecting a category or region updates related charts.
- Apply a consistent theme, fonts, labels, and spacing.
- Add dashboard navigation buttons if multiple dashboards are created.

7. Tableau Calculated Fields

```
// Profit Margin
SUM([Profit]) / SUM([Sales])

// Average Order Value
SUM([Sales]) / COUNTD([Order ID])

// Sales Growth
(SUM([Sales]) - LOOKUP(SUM([Sales]), -1))
/
ABS(LOOKUP(SUM([Sales]), -1))
```

8. Recommended Visuals

Business Question	Visual	Fields	Reason
How much did we sell?	Card	Total Sales	Fast KPI summary
How profitable are we?	Card	Total Profit / Margin	Profitability overview
Are sales increasing?	Line Chart	Date + Sales	Best for trends
Which category performs best?	Bar Chart	Category + Sales/Profit	Easy comparison
Which region performs best?	Column/Bar Chart	Region + Sales/Profit	Regional comparison
Which products lead?	Table	Product + Sales + Profit	Detailed ranking
How do filters affect results?	Slicers	Date/Region/Category/Segment	Interactivity

9. Interactivity Features

- Date slicer: Select year, quarter, month, or custom date range.
- Region slicer: Compare geographical performance.
- Category slicer: Focus on specific product categories.
- Segment slicer: Analyze customer segments where available.
- Cross-filtering: Selecting a chart element should update related visuals.
- Tooltips: Display additional Sales, Profit, Quantity, and Margin information.
- Drill-down: Allow users to move from Year → Quarter → Month where appropriate.
- Navigation: Use buttons to move between Overview and detailed analysis pages.

10. Consistent Design Theme

- Use one professional color palette throughout the dashboard.
- Use the same font family and similar font sizes across visuals.
- Keep KPI cards at the top and detailed charts below.
- Avoid unnecessary 3D charts and excessive decoration.
- Use clear titles such as 'Monthly Sales Trend' instead of vague titles.
- Show units and currency consistently.
- Use conditional formatting to highlight positive and negative performance.
- Leave sufficient white space so the dashboard is easy to read.

11. Handling Large Datasets

For large datasets, performance becomes important. In Power BI, use a proper star schema, remove unused columns, prefer measures over unnecessary calculated columns, use Import mode when appropriate, and use incremental refresh for large time-based datasets. In Tableau, use extracts where appropriate, reduce unnecessary fields, use filters carefully, and optimize data sources and calculations.

12. Business Insights to Look For

- Identify the month or period with the highest sales.
- Identify the most profitable category and sub-category.
- Find regions with high sales but low profit.
- Identify products with high sales but weak profitability.
- Compare current sales with the previous period/year.
- Find whether growth is consistent or driven by a small number of periods.
- Identify segments or customer groups generating the most revenue.

13. PPT Summary Structure

The required PPT summary can contain 6 slides:

Slide 1 – Title: Task 4: Dashboard Design | Sales & Financial Performance Dashboard | Student name

Slide 2 – Objective & Dataset: Objective, dataset source, tools, important columns

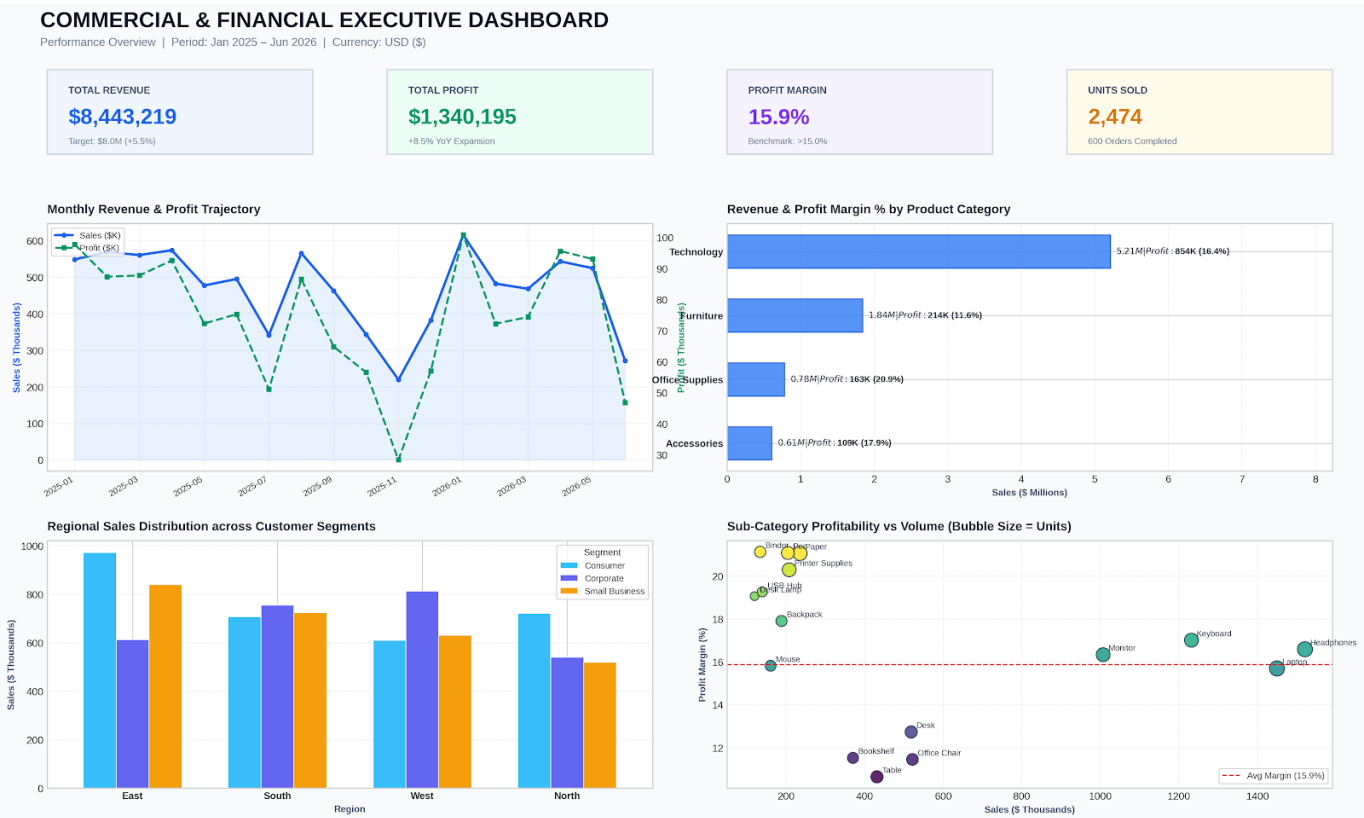
Slide 3 – Dashboard KPIs: Total Sales, Profit, Quantity, Profit Margin, Growth

Slide 4 – Dashboard Design: Insert screenshot of the completed Power BI/Tableau dashboard

Slide 5 – Key Insights: Top category, top region, sales trend, profit trend, major business finding

Slide 6 – Conclusion: How the dashboard supports business decisions and what was learned

15. Dashboard Screenshot Section



16. Interview Questions and Answers

1. What are the key elements of a dashboard?

The key elements are KPIs, charts and graphs, filters or slicers, clear titles, summary cards, interactive controls, consistent formatting, and a layout that tells a clear business story. A good dashboard focuses on the information needed for decisions.

2. What is a KPI?

KPI stands for Key Performance Indicator. It is a measurable value used to evaluate business performance against a goal. Examples include Total Sales, Total Profit, Profit Margin, Customer Count, and Sales Growth %.

3. What are slicers in Power BI?

Slicers are visual filtering controls in Power BI. They allow users to select values such as date, region, category, or segment, and the connected dashboard visuals update according to the selection.

4. Difference between Power BI and Tableau?

Both are business intelligence and data visualization tools. Power BI is closely integrated with the Microsoft ecosystem and uses DAX for calculations. Tableau is widely known for flexible visual analytics and interactive exploration. The best choice depends on the organization's ecosystem, licensing, existing skills, data sources, and reporting requirements.

5. How do you make a dashboard interactive?

Use slicers/filters, cross-filtering, drill-down, tooltips, bookmarks or buttons, navigation, and appropriate visual interactions. The interaction should help users answer questions without making the dashboard confusing.

6. How do you deal with large datasets in dashboards?

Remove unnecessary columns and rows, clean data before loading, use a star schema, create efficient measures, optimize relationships, use aggregations where suitable, and use Power BI incremental refresh or Tableau extracts when appropriate. Performance should be checked with the tool's optimization features.

7. What chart types do you use for trend analysis?

Line charts are usually the best choice for time-series trends because they clearly show movement over time. Area charts can also be used, while column charts are useful for comparing discrete periods such as monthly sales.

8. Why are dashboards useful for business decisions?

Dashboards combine important metrics and visual analysis in one place. They help stakeholders identify trends, compare performance, spot problems, monitor KPIs, and make faster data-driven decisions.

17. Conclusion

The Sales and Financial Performance Dashboard provides an interactive way to monitor important business metrics. By combining KPI cards, slicers, time-series charts, category and regional comparisons, and detailed tables, stakeholders can quickly understand business performance. The task demonstrates how Power BI or Tableau can transform structured sales data into useful information for data-driven decision making.